Introduction

In this chapter, you will:

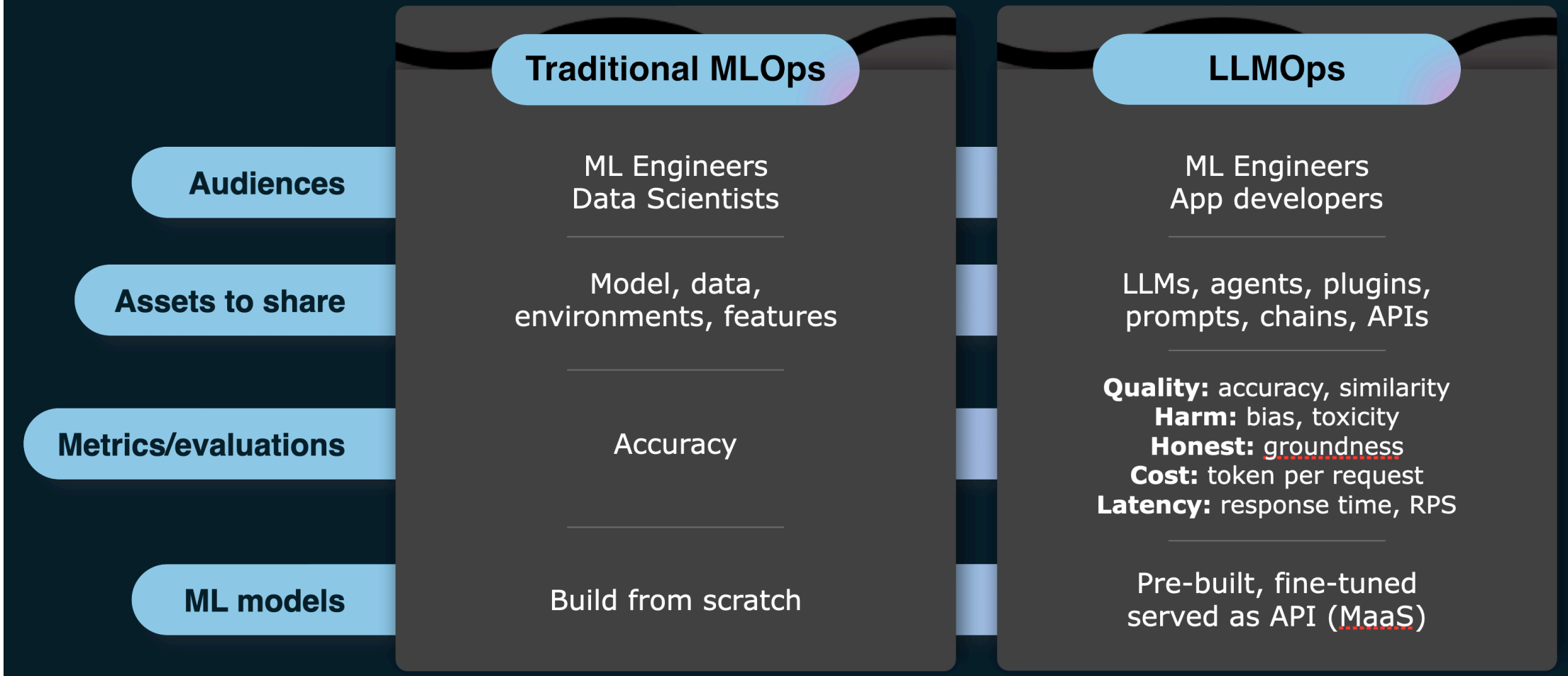
- Understand the Paradigm Shift from MLOps to LLMOps
- The LLM Lifecycle
- Lifecycle Tooling
- Lifecycle Metrification and Evaluation

Understand the Paradigm Shift from MLOps to LLMOps

LLMs are a new tool in the Artificial Intelligence arsenal, they are incredibly powerful in analysis and generation tasks for applications, however this power has some consequences in how we streamline AI and Classic Machine Learning tasks.

With this, we need a new Paradigm to adapt this tool in a dynamic, with the correct incentives. We can categorize older AI apps as "ML Apps" and newer AI Apps as "GenAI Apps" or just "AI Apps", reflecting the mainstream technology and techniques used at the time. This shifts our narrative in multiple ways, look at the following comparison.

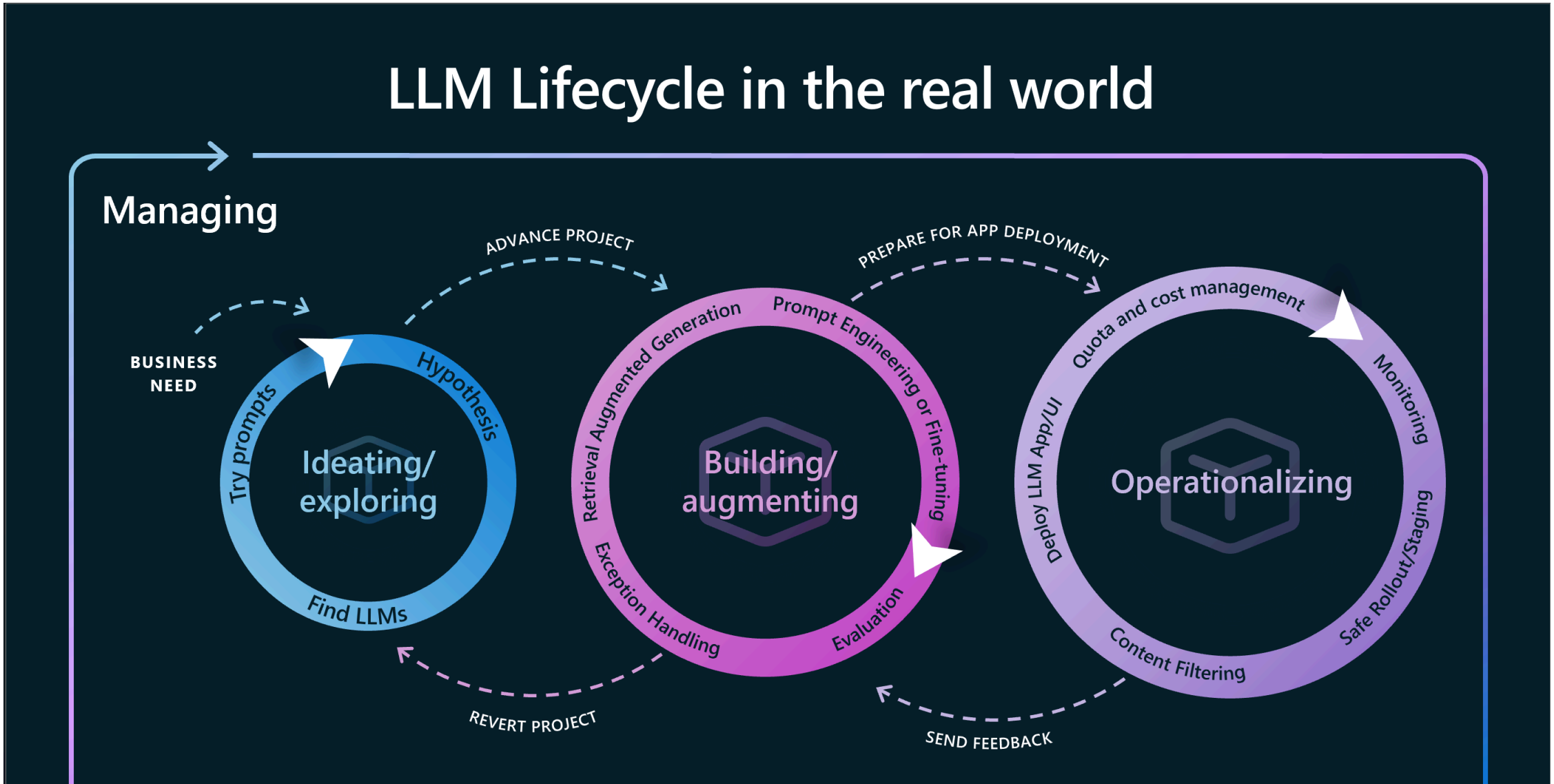
The paradigm shift—from MLOps to LLMOps

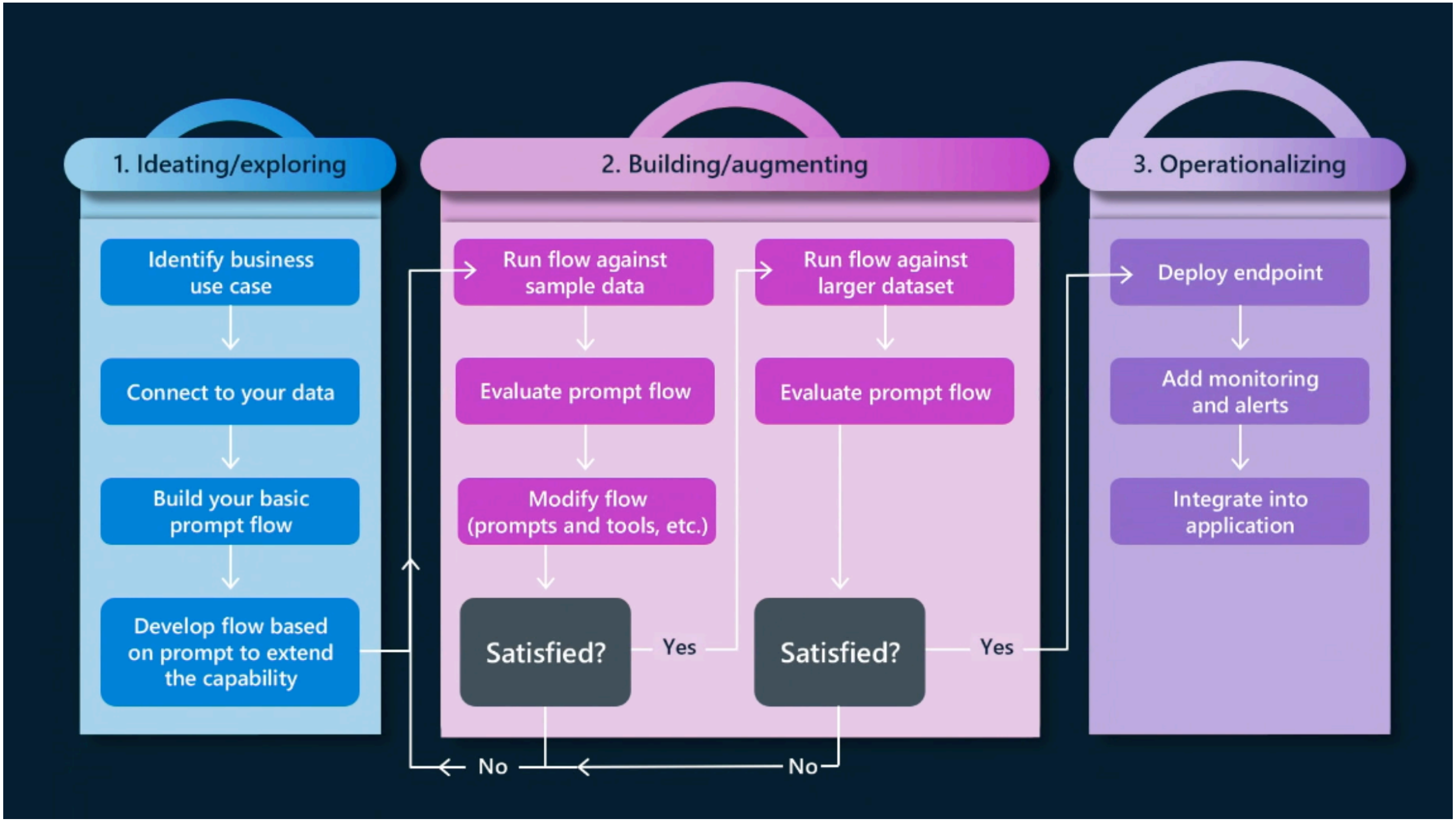


Notice that in LLMOps, we are more focused on the App Developers, using integrations as a key point, using "Models-as-a-Service" and thinking in the following points for metrics.

- Quality: Response quality
- Harm: Responsible AI
- Honesty: Response groundedness (Makes sense? It is correct?)
- Cost: Solution Budget
- Latency: Avg. time for token response

The LLM Lifecycle



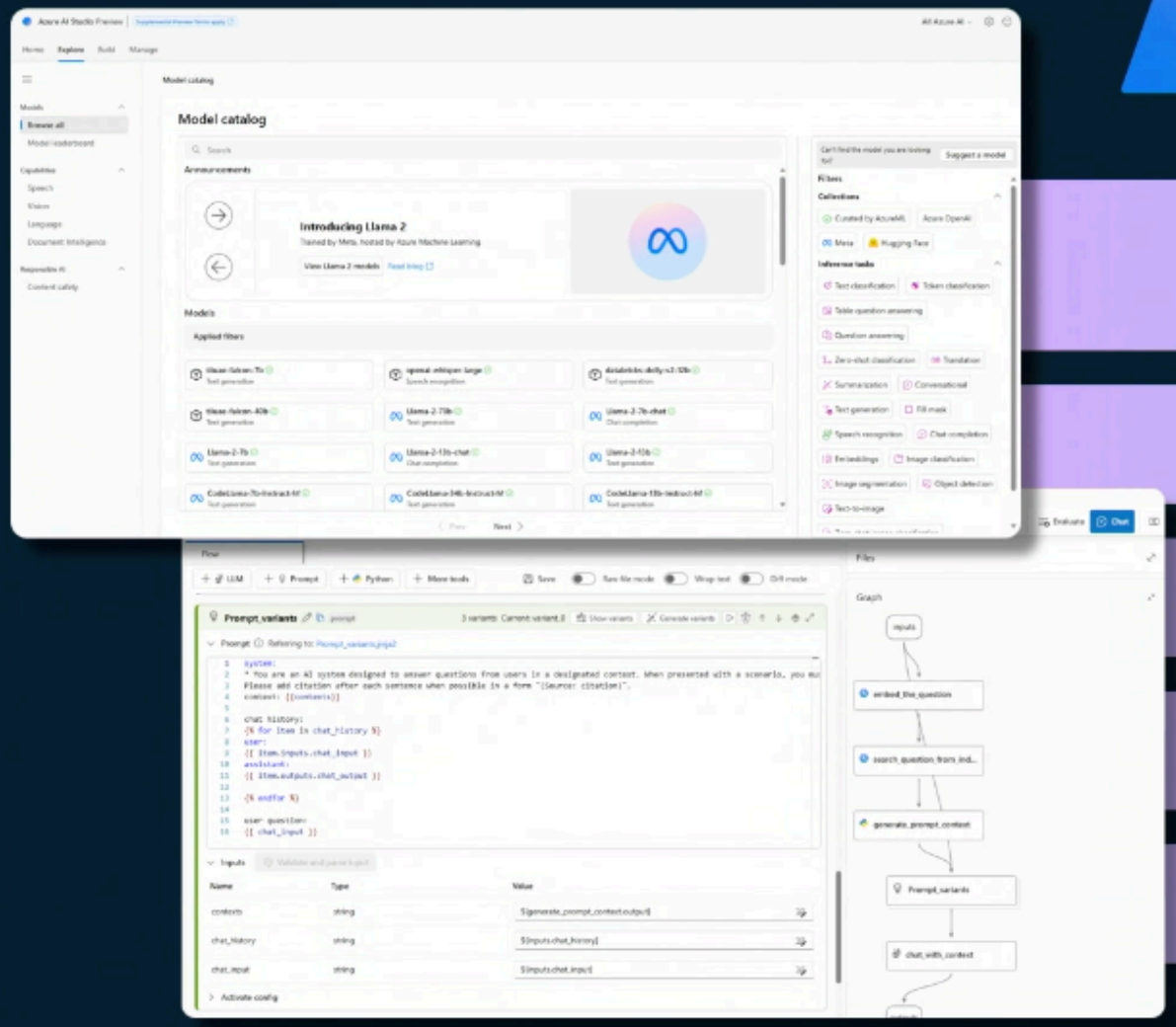

1. Ideating/exploring
Identify business use case
Connect to your data
Build your basic prompt flow
Develop flow based on prompt to extend the capability
2. Building/augmenting
Run flow against sample data
Evaluate prompt flow
Modify flow (prompts and tools, etc.)
Satisfied?
Yes
No
Run flow against larger dataset
Evaluate prompt flow
Satisfied?
Yes
No
3. Operationalizing
Deploy endpoint
Add monitoring and alerts
Integrate into application

Lifecycle Tooling

For Tooling, Microsoft provides the Azure AI Platform and PromptFlow facilitate and make your cycle easy to implement and ready to go.

The Azure AI Platform, allows you to use Microsoft Foundry. Microsoft Foundry (formerly Azure AI Studio) is a web portal that lets you explore models, samples and tools, manage your resources, and use UI development flows as well as SDK/CLI options for Code-First development.

Azure AI is a platform for Generative AI



- Access to thousands of LLMs from OpenAI, Meta, Hugging Face
- Data grounding with RAG
- Prompt engineering/evaluation
- Built-in safety and responsible AI
- Continuous monitoring for LLMs

Simplifying LLM Ops with Azure AI Platform

Azure AI Resource	Azure AI Services	Azure AI Project	Azure AI Search	Azure CosmosDB
				
Manage Ops including billing, permissions, policies, compute, service access	Built-in capabilities you can activate. Use default Open AI and Content Safety services	Build Workspace to organize work & save state. Use Prompt Flow, Filters & Deployments	Vector Search required for RAG. Add indexes for your product data for efficient query	Managed NoSQL database for app data at scale. Use it for customer id and order history

Construct, from Proof-of-Concept(POC) until large scale applications with PromptFlow:

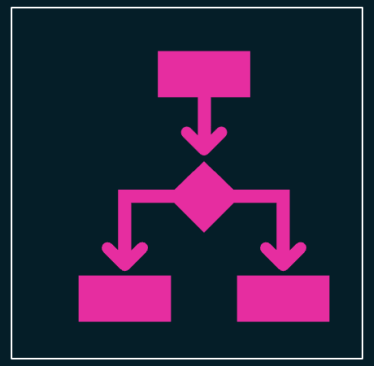
Streamlining LLM App Dev with Prompt Flow

VS Code Extension



Design & Build your LLM App as a DAG with inputs, nodes, outputs

Visual DAG Functions



Extend & Run your prompt flow with visual & function tools

Evaluation Metrics



Test & Tune your prompt flow for quality and responsible AI

Cloud Deployment



Push & Deploy your prompt flow to Azure for app integrations

Azure AI Studio UI



Integrate & Iterate with cloud hosted runtime & LLM app endpoint